A Cross-Linguistic (Chinese-English) Study of Agent Adaptation under Content Moderation

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ABSTRACT

Policy-oriented LLM agent simulations often treat language as a neutral channel, but communication under regulation may depend on the linguistic system itself. We study content moderation as a policy intervention and ask how language form might change its downstream effects. In a Chinese–English comparison with matched tasks and calibrated enforcement, we examine how writing-system affordances shape indirect expression and how agents adapt over repeated interactions. We focus on governance-relevant outcomes, including coordination cost, convention formation, and information recovery accuracy under constraint.

CCS CONCEPTS

• Computing methodologies → Multi-agent systems

KEYWORDS

Large Language Models (LLMS), Agent-based simulation, content moderation

1 Introduction and Related Work

Moderation constraints can reshape how groups communicate, coordinate, and complete shared tasks. Treating content moderation as an institutional constraint, we examine whether language form—particularly the writing system—causally structures agents' adaptive strategies when policies are held fixed. Observational studies have shown that communities frequently adopt euphemistic 'code words' to mask intent and bypass automated filters [8], yet isolating why these patterns emerge is difficult in real-world platforms like Reddit, where community rules and contexts are tightly entangled [7].

Recent work has demonstrated that multi-agent communication games can reproduce complex linguistic phenomena [4], and LLM- based systems can effectively simulate the evolution of coded language [1] and the emergence of community-specific protocols [8]. Complementing this, research into agent societies has shown that such systems can spontaneously generate and adhere to complex social norms through natural language discourse [3]. However, while these simulations have explored general mechanisms of language, dissemination dynamics [6], and norm evolution [1, 3], there remains a gap in understanding how specific linguistic structures—such as the logographic nature of Chinese versus the alphabetic system of English—differentially shape these adaptive trajectories, particularly as LLMs do not represent or process all languages equally [5] and exhibit significantly higher safety risks in non-English contexts [10].

A Chinese–English comparison provides a useful lens. Both communities develop indirect expression under pressure, yet they do so through different affordances: Chinese enables character-level transformations, while English relies more on meaning-level reformulations, such as the development of euphemistic lexicons that repurpose benign vocabulary [8]. These differences may affect how quickly workarounds stabilize into shared conventions [3, 4], the amount of repair required, and overall task success.

Although LLM-based systems have established the potential for creating believable interactive simulacra [2], many studies treat language as a neutral medium. Furthermore, while empirical research has identified the challenges of detecting implicit evasion and rule-based moderation [7, 8], it remains difficult to isolate the causal role of the writing system itself. Given that LLMs exhibit inherent cross-lingual disparities [5], we propose a matched setup utilizing generative agent architectures [2] to track the full arc of adaptation—from invention to normative stabilization [3, 4].

2 Framework Design

To study how restricted speech may reshape interaction across languages, we use a regulated communication setup where moderation is applied online during agent–agent dialogue. We aim to keep the task and restriction regime comparable across conditions and treat language of interaction (Chinese vs. English) as the primary variable. The system includes two LLM-driven roles: Participant agents, which attempt to convey target information while staying compliant, and a Supervisor, which enforces the policy and flags potential violations. By placing enforcement inside the interaction loop, moderation pressure becomes observable as agents try to complete an information- transfer task under constraint.

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Participant agents are modular and maintain persistent components for repeated interaction, including dialogue state, memory, and lightweight reflection. The Supervisor acts as an LLM-based moderation module that evaluates each turn for compliance. When the Supervisor detects a violation, it can interrupt the exchange and provide brief feedback, creating a signal that participants may use to adjust future phrasing. To approximate common review pipelines while keeping enforcement reproducible and tunable, we use a two-stage process: a surface-level screen followed by an LLM-based review for turns that pass the first stage. We calibrate Supervisor prompts and thresholds on matched development tasks to keep intervention behavior broadly comparable across languages.

To support adaptation over time, the framework maintains explicit records. A Dialogue History preserves within-round conversation for assessing whether target information was successfully transmitted. A cumulative Violation Log records flagged turns and Supervisor feedback across rounds, providing persistent signals that can influence later behavior and support the emergence of shared conventions.

Each run follows a repeated cycle of interaction, enforcement, and adjustment. We initialize roles, background knowledge, and task definitions, then allow participants to dialogue under constraint while the Supervisor checks each turn and intervenes when needed. Adaptation is operationalized through two reflection steps: before each round, agents review the Violation Log to infer likely triggers and update communication heuristics; after each round, they review the Dialogue History to assess task success and summarize lessons learned. A short planning step consolidates updates for the next round. The loop repeats for a fixed number of rounds or until key behaviors become relatively stable.

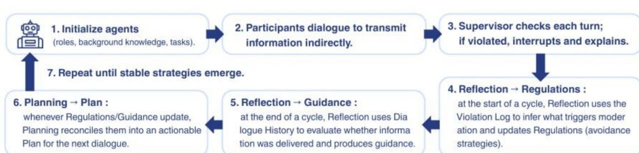

Figure 1: The Iterative Adaptation Framework for Regulated Multi-Agent Interaction. The diagram illustrates the closed-loop process where agents attempt information transfer under moderation, reflect on violation logs and dialogue history, and update their communication heuristics across repeated cycles.

3 Chinese–English Matching Strategy

To study how language form may affect adaptation under content moderation, we use meaning-based matching rather than word-for- word translation. The goal is to make the Chinese and English conditions comparable in task demand and enforcement behavior, so differences are less likely to be driven by uneven task difficulty or moderation strictness and can be interpreted more directly in terms of writing-system affordances.

3.1 Meaning-Based Task Alignment

We match tasks by intent and information requirements, not by identical prompts. Each task is defined by a small set of required information items (e.g., who/what/when/where/relations) and a shared success rule: the receiver should recover these items with low ambiguity. We also keep interaction budgets comparable across languages (e.g., similar turn or token limits) so both conditions face similar pressure on clarification and repair.

3.2 Policy Calibration and Supervision

We calibrate the Supervisor to keep enforcement broadly comparable across languages. Moderation follows a two-stage process. First, we align restricted content at the concept level and build language-specific keyword lists with similar coverage of common variants. Second, a centralized LLM-based Supervisor applies the same review rubric to detect indirect mentions that pass the surface screen. We run calibration trials using a small held-out set of compliant and violating messages in each language, then adjust the review criteria until enforcement behavior is broadly similar (e.g., comparable trigger rates and a similar balance of false positives and false negatives).

3.3 Metrics for Cross-Linguistic Comparison

Across repeated cycles, we track how indirect strategies and conventions develop using a small set of measures. We measure time to stability (how many cycles until behavior becomes relatively stable with low violations), strategy diversity (how varied the strategy categories and reformulations are over time, without assuming a specific strategy in advance), and transmission fidelity (how accurately the receiver recovers the intended information items). We also track coordination cost using repair and clarification signals (e.g., number of repair turns) to capture the effort required to stay both compliant and effective.

This matching strategy supports interpretable Chinese–English comparisons under a shared restriction regime while keeping the design flexible before final task scenarios are fixed.

4 Research Questions and Hypotheses

This study explores whether language form affects how agents adapt under the same moderation pressure. With matched tasks and calibrated enforcement, we compare Chinese and English and look at what changes over repeated rounds. We focus on three outcomes that matter for policy questions: how much effort it takes to coordinate under constraint, what shared ways of speaking (conventions) emerge, and whether the task can still be completed accurately. RQ1: Under similar moderation, do Chinese and English groups differ in how fast their behavior becomes stable (e.g., fewer violations and less strategy switching over rounds)? RQ2: Do the two languages tend to develop different kinds of indirect expression over time (e.g., meaning-level rewrites vs. character/orthography-based changes)? RQ3: With the same interaction budget, how do they balance coordination effort (repairs and clarifications) and task success (information recovery accuracy)? Chinese groups may stabilize sooner if character-level changes help people converge on shared patterns, though strict semantic review might reduce this effect. The strategy mix may also differ, with English leaning more on paraphrase/metaphor and Chinese showing more character-level or coded forms. Even when accuracy is similar, the amount of repair and back-and-forth may differ across languages.

5 Discussion

This work is primarily about what we might learn from cross-linguistic agent simulations under moderation pressure. Language may shape how groups adapt, not only in what they say, but also in how quickly shared conventions form and how much effort coordination requires [3, 4]. If we run simulations in only one language, some observed patterns might not generalize. We therefore use matched tasks and calibrated enforcement mainly to make Chinese–English comparisons more interpretable, so differences are less likely to come from task difficulty or uneven strictness. As observed in empirical studies of platform-scale moderation, rules are often community-specific and difficult to generalize across languages [7], justifying our controlled simulation approach.

Because this topic can be sensitive, we focus on describing high-level behavioral patterns and system-level outcomes rather than actionable "how-to" details. We report strategy categories and interaction dynamics instead of concrete substitution strings, and we can use sandboxed or synthetic placeholders to avoid direct mapping to real prohibited content. We also treat model bias as a practical concern, acknowledging that LLMs do not represent or process all languages with equal efficacy [5]. Furthermore, many current safety frameworks exhibit crucial vulnerabilities in multilingual contexts, particularly in defending against adversarial attacks using low-resource languages or code-switching [9]. Therefore, we plan robustness checks by varying models, prompts, and enforcement settings.

Finally, we expect moderation and behavior to co-evolve: enforcement can shape how agents communicate, and new communication forms may change what enforcement catches or misses. This dynamic underscores the need for more transparent and robust multilingual guard agents capable of reasoning across diverse linguistic contexts [9]. This iterative adaptation echoes the evolutionary trajectories of 'coded language' [1] and the complex dissemination dynamics observed in recent social AI simulations [6].

We also note limitations. Outcomes may depend on the underlying LLM, and equal strictness across languages is difficult to guarantee; calibration reduces mismatch but cannot remove it. Our tasks are structured for measurement, while real-world communication is richer, so the simulation is best viewed as a way to surface mechanisms and patterns within a generative agent architecture [2] rather than predict any specific platform.

6 Conclusion

We present an exploratory study plan that uses a controlled LLM-agent simulation to observe how moderation pressure might lead to different adaptation patterns in Chinese and English. By matching tasks and adjusting enforcement levels, we aim to make cross-linguistic differences easier to understand, focusing our analysis on how new conventions form, the effort needed for coordination, and overall task success. Moving forward, we plan to run a small case study to track initial trends, before expanding to more complex interactions. We also hope to share a simple logging and calibration protocol to help others replicate this work.

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